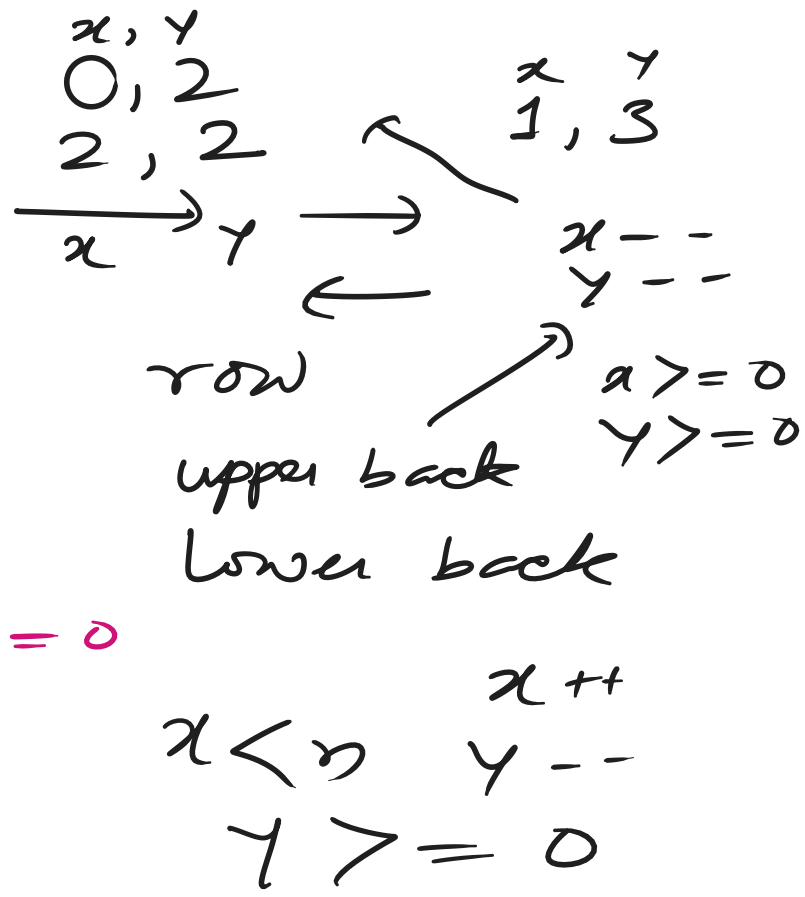
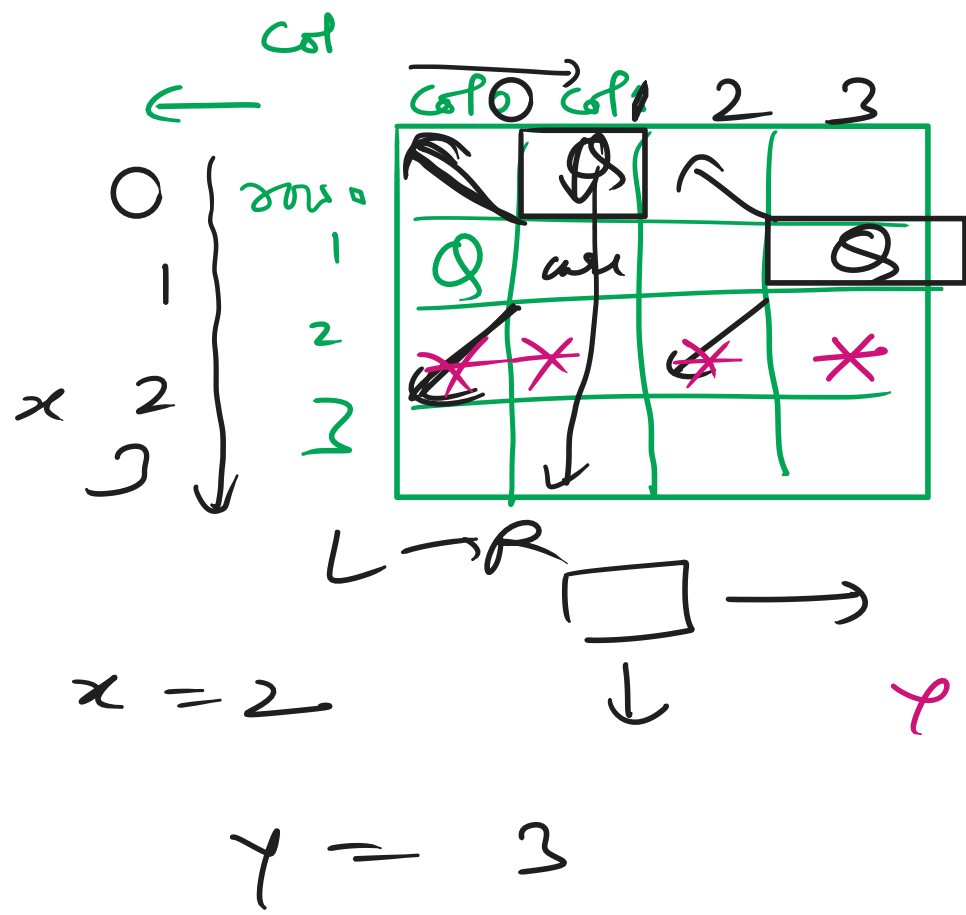
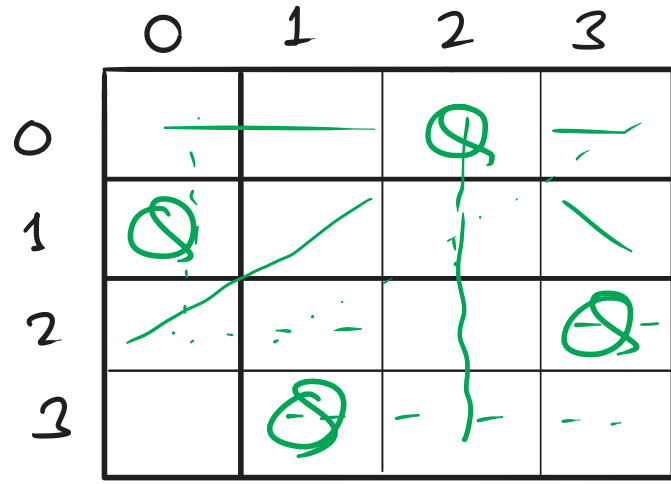
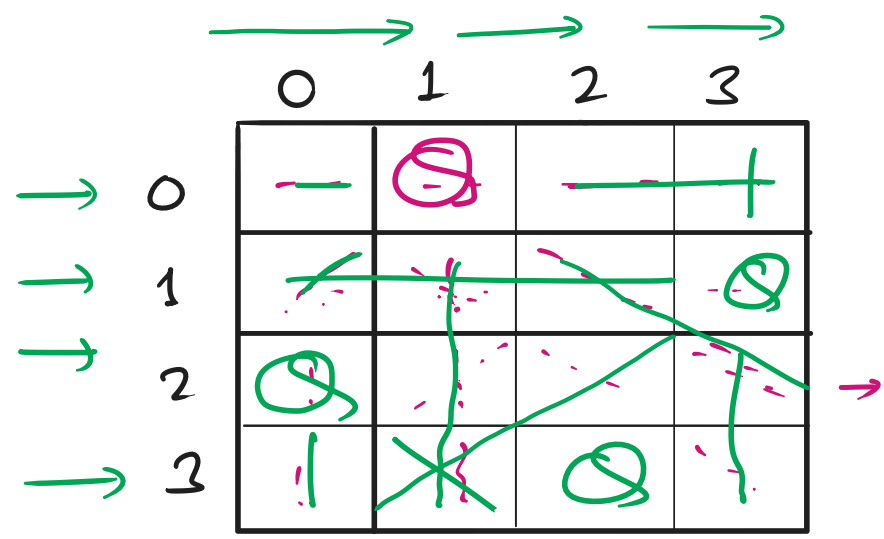


Backtracking Day 2 :

① The N Queens Problem

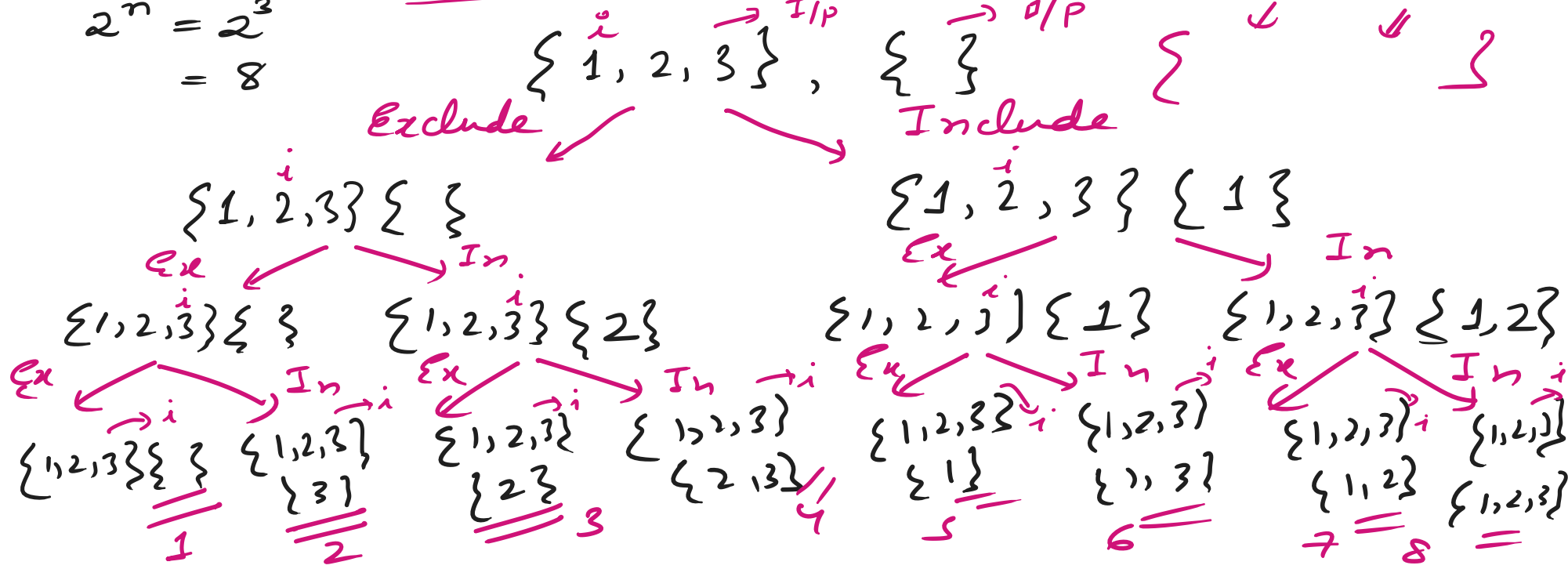
Row → 1 Q
Col → 1 Q



Power Set $\Rightarrow$ Set of all subsets of an array :

$$\{1, 2, 3\} = \{\{\}, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}\}$$

$$2^n = 2^3 = 8$$



Subsequences of a string $\Rightarrow$

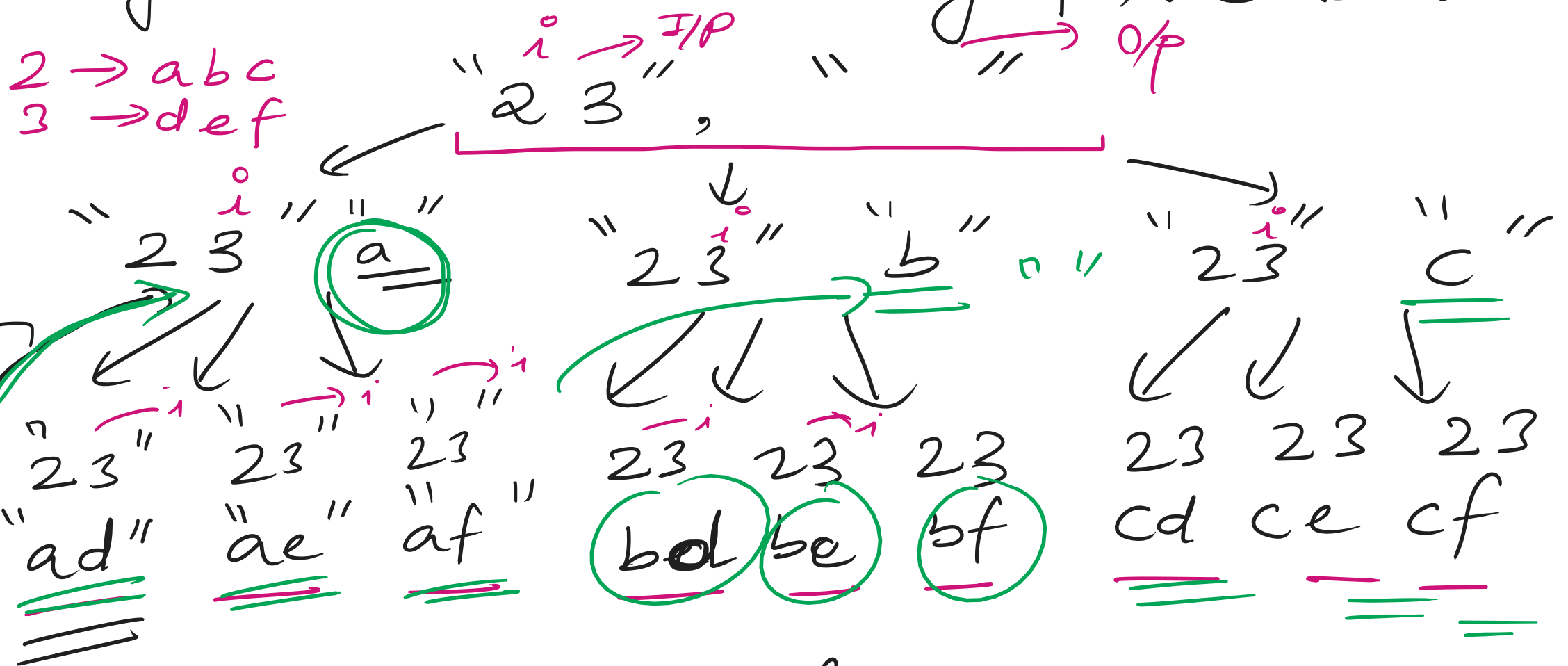
Not needed $\{abc\} \{ \}$
 $\{ \} \{a\} \{b\} \{c\} \{a, b\} \{a, c\} \{b, c\} \{a, b, c\}$

* Note: The empty string should not be displayed in the output.

[10] [0-9] Phone KeyPad Problem : BackTracking / Recursion



Backtracking



Remember that when all the possibilities of "a" have been considered, we have to remove "a" [Backtrack] and so on for "b" & "c" as well.